

DALEN WAGGA

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Portfolio: hypnotic8008.github.io/CatchSim | linkedin.com/in/dalen-wagga | github.com/Hypnotic8008

EDUCATION

University of Alabama in Huntsville — B.S. Aerospace Engineering

Huntsville, AL

GPA: 3.4 | *Sophomore Standing* | *Expected May 2029*

Relevant Coursework: Differential Equations, Statics, Principles of Aero/Astro, Numerical Methods, Engineering Economy, Calculus 2/3, Linear Algebra, Physics 1

PROFESSIONAL EXPERIENCE

Computational Fluid Dynamics Intern — McLaurin Aerospace

May 2026 – Present

Huntsville, AL

- Execute Cart3D CFD simulations on pre-built AIM-120 missile geometries across parametric angle-of-attack and roll-angle sweeps to characterize aerodynamic performance
- Built Python pipelines to automate batch simulation execution and post-processing, computing aerodynamic forces and moments (Fx, Fy, Fz, Mx, My, Mz), static margin, and minimum turn radius
- Run viscous CFD in Loci/CHEM across the full mesh-to-post-processing pipeline (AFLR3, gridmerge), tuning boundary-layer wall spacing for $y^+ < 1$
- Resolved HPC cluster and coordinate-transform issues (PBS/qsub job submission, aeroballistic coordinate conversions) to keep the simulation pipeline reliable

PROJECTS

Super Heavy Booster Catch Simulation — Personal Project

github.com/Hypnotic8008/CatchSim

- Built a 6-DOF quaternion-based Python sim of a Super Heavy V3 booster from hot-stage separation to tower catch, with closed-loop guidance through flip, boostback, grid-fin entry, and the landing burn
- 100-case Monte Carlo under 3σ dispersions (thrust, Isp, mass, drag, density, winds, separation state): **101/101 caught**, up from 1/41 with baseline guidance; median lateral error 0.15 m vs. 1 m limit
- Diagnosed each failure mode and redesigned guidance: closed-loop boostback targeting with in-flight thrust/mass-flow estimators, grid-fin entry steering, and a wind disturbance observer

Water Deluge System — Personal Project

Full write-up + figures on portfolio

- Designing a small-scale water deluge system modeled on the SpaceX Starship flame deflector; modeled a stainless-steel deluge plate with ~100 angled orifices (0° – 30°) in Autodesk Inventor
- Developed two meshing approaches in Ansys — an initial polyhedral mesh, then a ~1M-cell hybrid mesh (hex core with tet refinement at the orifice plate) — validated against element quality, orthogonal quality, and skewness before committing to a solve
- Specified a full bill of materials for an instrumented ground test — solenoid valve, nitrogen regulator and tank, pressurized water tank, and 3/4" piping — with vendor sourcing and cost tracked per line item

Starship Forward Flap — CFD & FSI Analysis — Personal Project

Full write-up + figures on portfolio

- Modeled a Starship forward flap in Inventor and ran CFD in Ansys Fluent (k- ω SST) at supersonic inflow, resolving leading-edge velocities to ~1,520 m/s and peak dynamic pressures near 1.4 MPa
- Post-processed in CFD-Post with velocity contours, surface streamlines, and pathlines to identify leading-edge acceleration and separated flow over the flap's upper surface
- Extracted a pressure-vs-velocity distribution across the flow field and mapped Mach 2, 25° AoA pressure loads onto an Ansys Static Structural model in a one-way FSI workflow

Tartarus Liquid Engine, Propulsion Sub-Team — Space Hardware Club

Jan 2026 – Present

- Modeled liquid-kerosene combustion behavior for a gaseous-only CFD tool by deriving temperature differences from the latent heat of vaporization, filling a gap in available literature
- Completed preliminary CAD for the Epimetheus injector, an upgrade over the team's Prometheus engine, using advanced surface/manifold modeling (projected geometry, work planes)

Rocketry Two Month, Mechanical Sub-Team — Space Hardware Club

Aug 2025 – Nov 2025

- Designed a coupler bulkhead and a removable, screw-in black-powder charge well (Inventor/Fusion) for an L2-class airbrake rocket, on an 11-person team under a 2-month design-to-manufacture timeline

ACTIVITIES

Space Hardware Club — Mechanical & Propulsion Sub-Teams | Alpha Tau Omega Fraternity | Sand Volleyball Club